Functions in Python

Defining Functions, Arguments, Return Values, Lambda Functions

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What are Functions?

- A block of reusable code that performs a specific task.
- Helps modularize code and improves readability.
- Reduces redundancy and makes debugging easier.
- Example:

```
def greet(name):
    return f"Hello, {name}!"

print(greet("Asif"))
```

Types of Functions

- 1. Built-in Functions (e.g., print(), len(), sum())
- 2. User-Defined Functions (Created using def keyword)
- Example:

```
print(square(5))  # Output: 25
```

Function Parameters and Return Values

- Parameters: Input values provided to a function.
- Return Statement: Sends back the function result.
- Default parameter: When no parameter is passed, a function can take default parameter value
- Example:

```
  def add(a, b):
      return a + b
print(add(3, 4))  # Output: 7
```

Defining Functions

- Use the 'def' keyword to define a function.
- Syntax:

  def function_name(parameters):

  # function body

  statements
- Example:

```
def greet():
    print("Hello, World!")
```

Function Arguments

- Functions can take arguments to work with data.
- Types:
  - Positional Arguments
  - Keyword Arguments
  - Default Arguments
  - Variable-length Arguments (*args, **kwargs)
- Example:

```
def greet(name):
    print(f"Hello, {name}!")
```

Return Values

- Use 'return' to send a result back to the caller.
- Syntax:

```
def add(a, b):
    return a + b
```

- Example:

```
result = add(5, 3)
print(result)  # Output: 8
```

Lambda Functions

- Anonymous, small, one-line functions using 'lambda'.
- Syntax: lambda arguments: expression
- Example:

  square = lambda x: x * x

  print(square(4)) # Output: 16

  Useful in map(), filter(), and sorting operations.

Lambda with filter()

- Example:

```
nums = [1, 2, 3, 4, 5]
evens = list(filter(lambda x: x % 2 == 0, nums))
print(evens)  # Output: [2, 4]
```

Summary

- - Functions help organize and reuse code.
- - Arguments pass data to functions.
- - Return values send results back.
- - Lambda functions are short, unnamed functions.